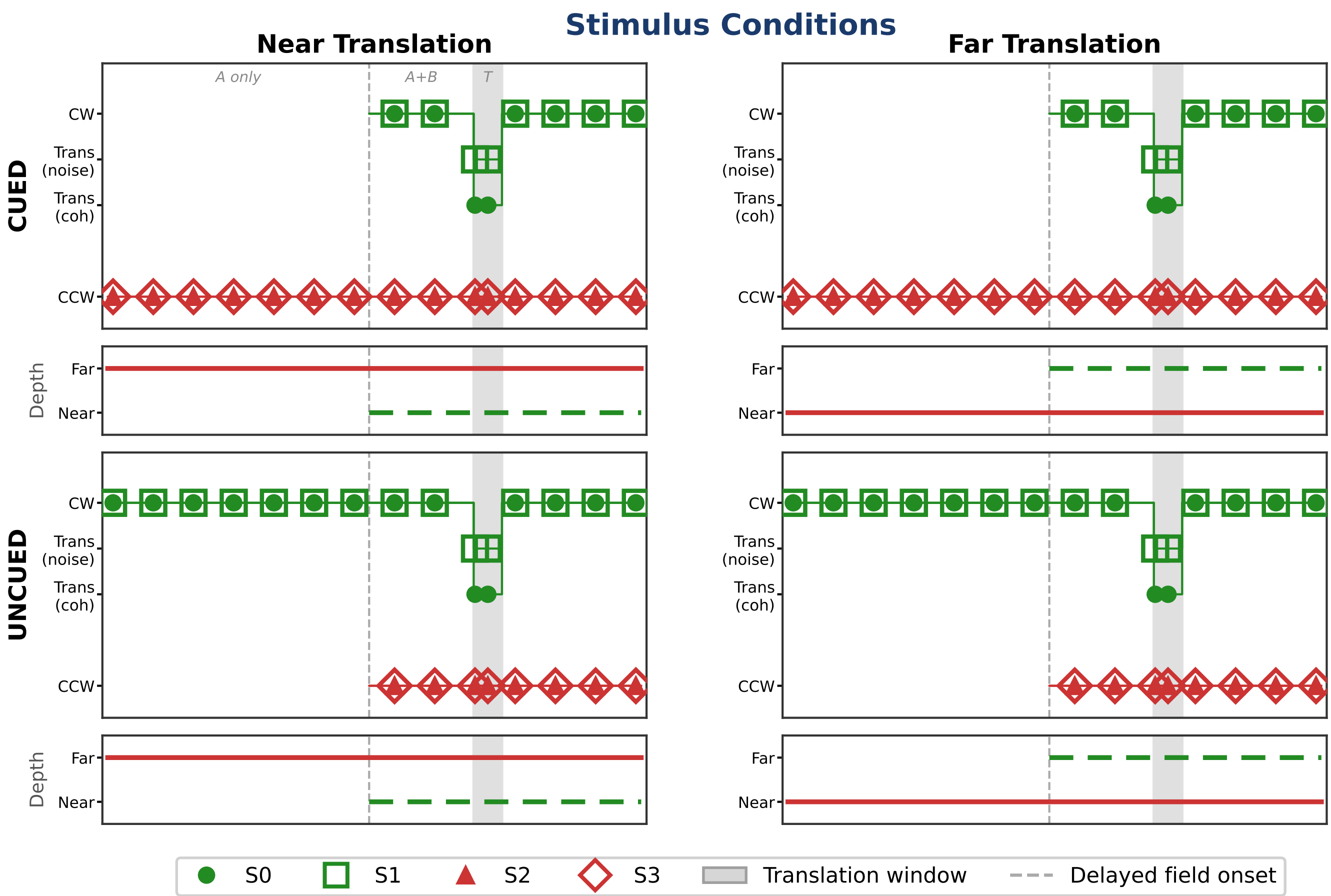
VRDots — Depth Baseline (No Swap)

Sessions: 260325_1831, 260325_1914 (0.10 m)   260325_2013 (0.03 m)

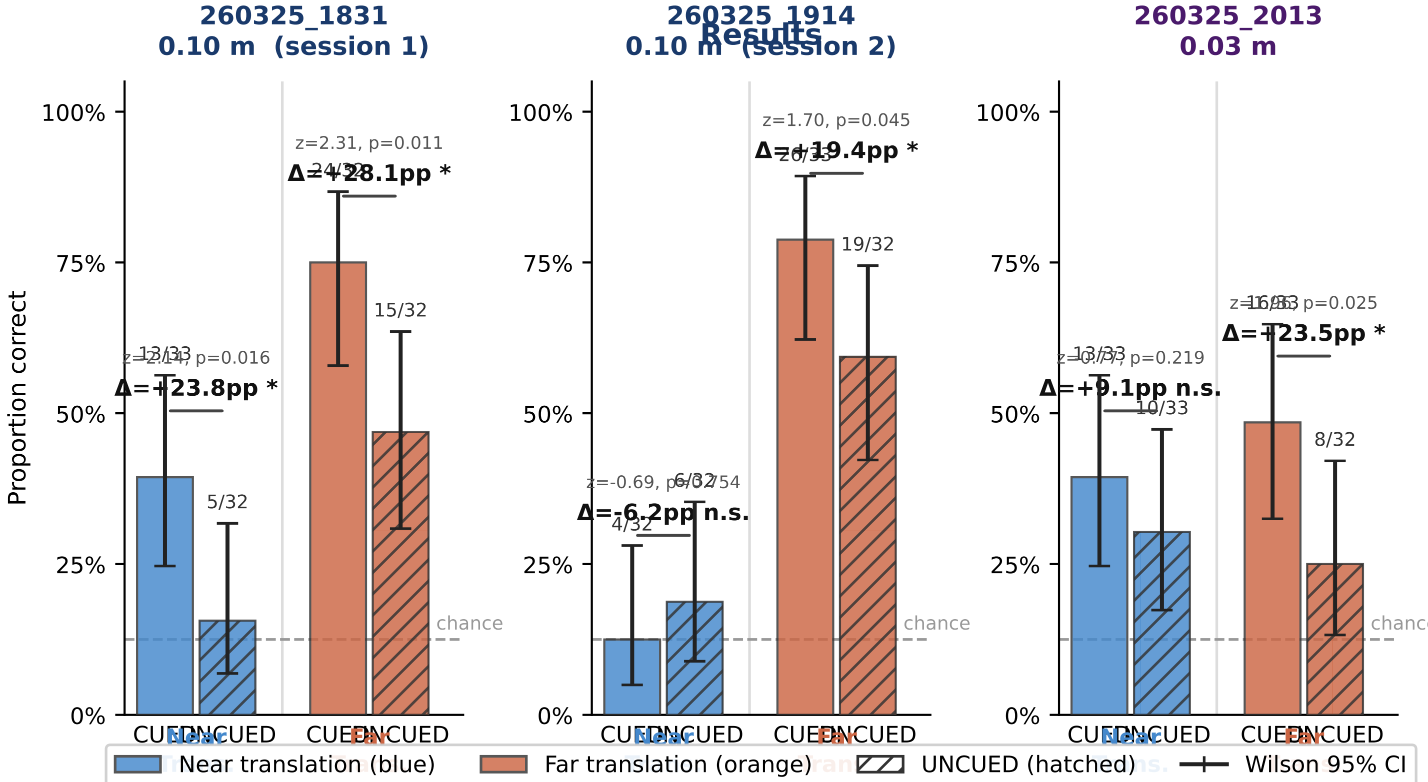
Three sessions introduced stereoscopic depth separation between the two rotating dot fields. The flat (no-depth, no-swap) design was otherwise unchanged: 750 ms delayed onset, 80 ms translation, 8-AFC direction judgment at 2.0 m, no swap. Sessions 260325_1831 and 260325_1914 used a depth separation of 0.10 m (fields at ±0.05 m from fixation depth), which pilot observation confirmed produced clearly visible stereo depth planes. Session 260325_2013 used 0.03 m separation, at which depth was weakly perceptible. Each session comprised 128–131 trials, fully crossing Cond (CUED/UNCUED) × DelayedFieldDepth (Near/Far) × DelayedFieldColor (G/R) × RotConfig (0/1) × TransDeg (8 directions), yielding approximately 8 trials per cell.

Trajectory diagrams show one representative color-rotation pairing. Columns are organized by the depth of the translating field (Near translation | Far translation) rather than by the depth of the delayed field. S0●/S1□ = initially CW field (green); S2▲/S3◇ = initially CCW field (red). Depth track (below each motion panel) shows which field occupies which plane; the CW/translating field is shown dashed to match its delayed onset in the CUED row.



Analysis: Near/Far labels throughout use translating-field depth. CUED Near = delayed field at Near (translates Near); UNCUED Near = non-delayed field at Near (translates Near, delayed field at Far). Proportion correct was computed per Cond × translating-depth cell, pooled across color, rotation, and translation direction (≈ 32 trials/cell). One-tailed z-test: CUED > UNCUED within each translating depth.

Results



Findings

Sessions 260325_1831 and 260325_1914 (0.10 m depth separation): a large performance asymmetry dominated both sessions. Far translation accuracy was high (≈75–79% CUED, ≈47–59% UNCUED), while Near translation accuracy was at or near chance (≈12–39% CUED, ≈16–19% UNCUED). This near-chance performance for Near translation is not a cueing failure but a stimulus-legibility floor: at 0.10 m separation the Near plane dots move in strong stereo depth, and their 2D translational signal is substantially masked or overridden by the large disparity change. The cueing effect (CUED > UNCUED) was significant for Far translation in both sessions (+28 pp, p=.011; +19 pp, p=.045) and significant for Near translation in session 1 (+24 pp, p=.016) but not session 2 (−6 pp, n.s.), where Near accuracy was too low to detect a cueing effect. Overall cueing was +26 pp (p=.002) in session 1 and +7 pp (n.s.) in session 2, reflecting the large influence of the near-chance Near cells.

Session 260325_2013 (0.03 m depth separation): with barely perceptible depth, both planes produced above-chance but modest accuracy. Far translation remained easier (CUED 49%, UNCUED 25%, Δ+24 pp, p=.025) and Near translation showed a smaller positive cueing effect (CUED 39%, UNCUED 30%, Δ+9 pp, n.s.). The overall cueing effect was +16 pp (p=.026). The residual Far > Near performance asymmetry at 0.03 m suggests that even weak stereo depth partially disrupts the legibility of Near-plane translation, though far less severely than at 0.10 m.

Notes

Notes: (1) The 0.10 m near-chance result for Near translation is a performance floor, not a cueing null. Any cueing effect that exists is undetectable when both CUED and UNCUED are near chance. This confound motivated the subsequent DepthParam experiment (0.03–0.15 m range) designed to find a depth separation that preserves legibility in both planes. (2) Session-to-session variability is high at n ≈32/cell. The reversal of the Near cueing effect between sessions 1 and 2 is likely noise. (3) The DepthBothPlanes session (260326_0550) and DepthSwap session (260326_0821) were also run at 0.10 m but are not included here as they introduce additional factors (swaps, joint-onset conditions) treated in later write-ups.